

COMPUTER HARDWARE TECHNOLOGY REVIEW QUESTIONS

A. MICROPROCESSORS

1. Give a diagram of a computer organization and explain the function of each block.
2. With a help of diagram, explain the basic three components of a processor.
3. How much RAM size would a 64-bits processor support?
4. Explain the factors influencing the performance of the processor.
5. The speed of the processor is measured inwhile the data transfer rate is measured in

B. MEMORIES

1. What is meant by primary memory, secondary memory and mass storage devices?
2. Define the terms storage capacity, memory word, storage location, and word length.
3. Define the following terms
 - (i) Access Time
 - (ii) Cycle time
 - (iii) Bandwidth
 - (iv) Volatile memories
4. Explain organization of tracks in magnetic memories with the help of figure.
5. What are optical memories?
6. Explain basic semiconductor memory organization with the help of figure.
7. With the help of figure explain the memory cell addressing.
8. Explain the memory cell organization. How a particular cell can be selected in this case?
9. A memory has access time of 50 n sec. It has an additional delay called precharge time of 20 n sec. Find out the access rate and bandwidth of this unit?
10. What is Dynamic RAM? Why they are most widely used?
11. What is Static RAM? Why they are most widely used?
12. Explain how a hard disk reads information it stores.
13. Provide advantages and disadvantages of a hard disk as a storage media

14. Explain how the CD-ROM stores data.
15. Provide advantages and disadvantages of a CD-ROM as a storage media

C. SYSEM BUSES AND BASIC CARDS

1. Define a computer system bus.
2. Explain the types of system buses based on the type of information they carry.
3. What is the difference between internal and external buses?
4. What do you understand by expansion bus?
5. With a block diagram explain the system bus related to computer operation.
6. Explain various expansion slots available on a computer motherboard.
7. Give the difference between ISA and EISA expansion slots.
8. Differentiate ISA from PCI expansion slots.
9. What do you understand by interrupt request lines in a computer?
10. Explain the function of DMA channels and I/O addresses.
11. Explain the term “Bus Mastering”.
12. Explain the factors you would base on while selecting a network interface card to Purchase.

D. POWER DEVICES

1. Give different DC voltage level produced by AT power supply with their use.
Specify also the colors of the power supply wires with their voltages.
2. What are the main differences etween AT and ATX power supplies?
3. What is an off-line UPS?
4. What is an Online UPS?

E. COMPUTER HARDWARE COMPONENTS

1. Which basic operating system function receives data from the keyboard?

- a) Input.
- b) Processing.
- c) Output.
- d) Storage.

2. In MSDOS, which of the following actions will result in a warm boot?

- a) Replacing the CPU.
- b) Pressing CTRL+ALT+DEL.
- c) Powering on the system.
- d) Regaining power after a power outage.

3. Which type of motherboard supports 3.3 volts?

- a) AT.
- b) ATX.

4. Which types of RAM is used for cache memory?

- a) SRAM(Static Random Access Memory).
- b) DRAM(Dynamic Random Access Memory).

5. Which type of storage devices uses a laser to record information?

- a) Hard disk.
- b) Floppy disk.
- c) CD-RW.
- d) Tape drive.

6. Which computer component can contain a dangerous amount of voltage.

- a) Motherboard.
- b) Monitor.
- c) Hard drive.
- d) CD ROM.

7. What is the function of the computer power supply?

- a) Convert DC to AC.
- b) Convert AC to DC.

- c) Eliminate spikes in the electricity.
- d) Eliminates brownout in the electricity.

8. Match the power supply to the voltage to the standard wire color.

	Power supply voltage	Wire color
Black	+12vdc	
Blue	-12vdc	
Orange	+5vdc	
Red	-5vdc	
White	+3.5vdc	
yellow	Ground	

9. What is the result of installing the LED connectors to the motherboard incorrectly?

- a) The motherboard will short out.
- b) LED status light will still work.
- c) LED status light will not work.
- d) The computer will beep constantly when powered on.

10. How are the hard drive and CD-ROM configured to be master and slave?

- a) CMOS setup utility.
- b) During partitioning.
- c) Set jumpers.
- d) During formatting.

11. How many volts do disk drive motors require?

- a) 3.3 volts.
- b) 5volts.
- c) 12 volts.
- d) 120 volts.

12. Which colored wire from a computer's power supply is the ground?

- a) Red.
- b) Black.

- c) Yellow.
- d) Blue.

13. What is the purpose of the heat sink?

- a) To cool the computer processor.
- b) To set the processor voltage.
- c) To set the processor speed.
- d) To ground the processor.

14. Some keys which allow you to access the CMOS Setup screen are DEL CTRL+ALT+DELETE and F2

- a) TRUE
- b) FALSE

15. Discuss on how the CD-ROM reads data from the CD.

16. Explain how the Hard disk drive writes and reads data to and from platters.

17 Explain the steps performed during the computer system power on self test (POST).

18. How can you identify an ATX motherboard?

19. Give the different types of expansion slots available on the Computer Motherboard.

20. Discuss why you would choose to build your own PC rather than buy a ready-made PC. You should list at least seven reasons to build your own PC and three reasons to buy a ready-made PC.

21. What are the reasons for partitioning a hard disk before installing an operating system in the computer?

22. Why is it necessary to format a drive before using a storage device?

23. What are the functions of the standoffs that sit on the motherboard?

24. Give the causes of the processor failure and how to trouble shoot this failure.

25. What are the functions of the data bus on the motherboard? What are some of the bus sizes that you can have on a motherboard?

26. Give the steps of Hard disk drive configuration for installing in your personal computer.

F: OPERATING SYSTEMS

1. Define the term “Operating system”

2. Give and explain three components of any operating system.
3. Give and explain the Functions of an Operating system .
4. Explain the following terms as related to operating system
 - i. Multiuser Operating system
 - ii. Multitasking Operating system
 - iii. Multiprocessing Operating system
 - iv. Multithreading Operating system
5. What are the three distinct sections that make up the disk operating system? Explain the function of each section.
6. Give and explain the common attributes for DOS files.